

Question 1: MINIMUM NO. OF JUMPS

Aim:

To find the minimum number of jumps required to reach the last index of an array using a Greedy approach.

Algorithm:

1. Start from the first index.
2. Maintain variables maxReach, steps, and jumps.
3. Update the farthest reachable index at each step.
4. When steps become zero, increment jumps and update steps.
5. Continue until the last index is reached.

Pseudocode:

```
Input n, arr[]
If n <= 1 return 0
Initialize maxReach = arr[0], steps = arr[0], jumps = 1
For i from 1 to n-1:
    If i == n-1 return jumps
    maxReach = max(maxReach, i + arr[i])
    steps = steps - 1
    If steps == 0:
        jumps = jumps + 1
        steps = maxReach - i
Return jumps
```

Python Program:

```
def min_jumps(arr):

    n = len(arr)

    if n <= 1:

        return 0

    maxReach = arr[0]

    steps = arr[0]

    jumps = 1

    for i in range(1, n):
```

```
    if i == n - 1:
        return jumps
    maxReach = max(maxReach, i + arr[i])
    steps -= 1
    if steps == 0:
        jumps += 1
        steps = maxReach - i
    return jumps
arr = [2, 3, 1, 1, 4]
print("Min Jumps =", min_jumps(arr))
```

Sample Input:

```
n = 5
arr = [2, 3, 1, 1, 4]
```

Sample Output:

Min Jumps = 2

Time Complexity: $O(n)$

Space Complexity: $O(1)$

Result:

The minimum number of jumps required to reach the last index was successfully found using a Greedy approach.